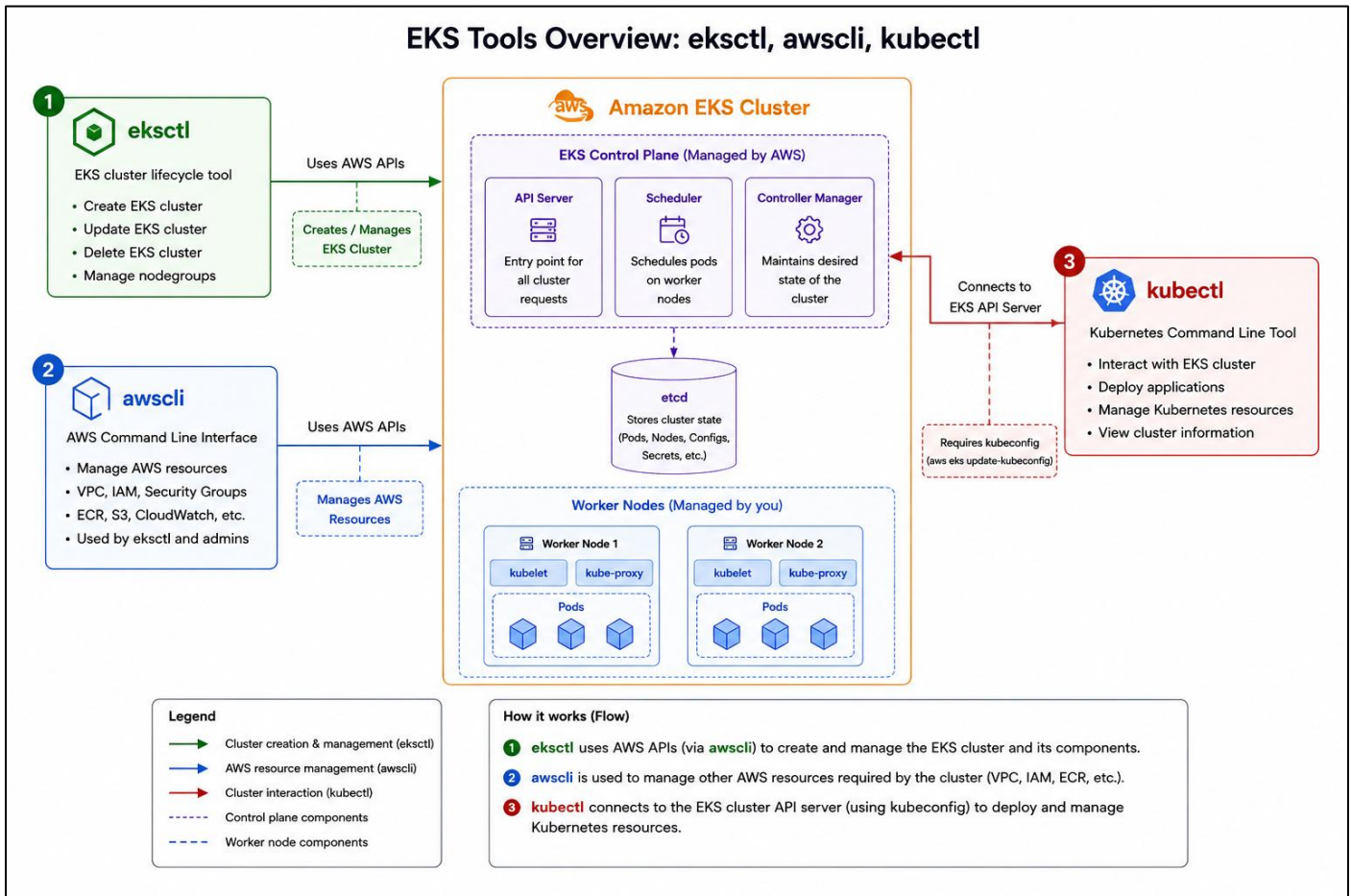


Kubernetes Day 2



Practical

1. In GitBash checking the versions

```
MINGW64:/c/Users/Bhavana/OneDrive/Desktop
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/OneDrive/Desktop
$ aws --version
aws-cli/2.34.53 Python/3.14.5 Windows/11 exe/AMD64
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/OneDrive/Desktop
$ eksctl version
0.226.0
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/OneDrive/Desktop
$ kubectl version --client
Client Version: v1.35.3-eks-bbe087e
Kustomize Version: v5.7.1
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/OneDrive/Desktop
$ kubectl version
Client Version: v1.35.3-eks-bbe087e
Kustomize Version: v5.7.1
Unable to connect to the server: dial tcp [::1]:8080: connect: No connection could be made because the target machine actively refused it.
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/OneDrive/Desktop
$ |
```

2. Configuring aws access key and secret access key

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/OneDrive/Desktop
$ aws configure
```

Tip: You can deliver temporary credentials to the AWS CLI using your AWS Console session by running the command 'aws login'.

```
AWS Access Key ID [None]: AKIAWV7SGBK6VZL2VURO
AWS Secret Access Key [None]: GJd7JbmUf8jiesOebKVs4sEQIYx8PRg4HIHsvcyL
Default region name [None]: us-east-1
Default output format [None]:
```

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/OneDrive/Desktop
$ aws sts get-caller-identity
```

```
{
  "UserId": "459532536509",
  "Account": "459532536509",
  "Arn": "arn:aws:iam:459532536509:root"
}
```

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/OneDrive/Desktop
$ |
```

3. Creating cluster, node group with instance and number of nodes

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/OneDrive/Desktop
```

```
$ eksctl create cluster --name b17d-eks-cluster --nodegroup-name b17d-nodegroup --node-type t3.micro --nodes 2 -
-managed
```

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/OneDrive/Desktop
$ eksctl create cluster --name b17d-eks-cluster --nodegroup-name b17d-nodegroup --node-type t3.micro --nodes 2 --managed
2026-05-27 20:59:55 [i] eksctl version 0.226.0
2026-05-27 20:59:55 [i] using region us-east-1
2026-05-27 20:59:57 [i] setting availability zones to [us-east-1c us-east-1b]
2026-05-27 20:59:57 [i] subnets for us-east-1c - public:192.168.0.0/19 private:192.168.64.0/19
2026-05-27 20:59:57 [i] subnets for us-east-1b - public:192.168.32.0/19 private:192.168.96.0/19
2026-05-27 20:59:57 [i] nodegroup "b17d-nodegroup" will use "" [AmazonLinux2023/1.34]
2026-05-27 20:59:57 [i] Auto Mode will be enabled by default in an upcoming release of eksctl. This means managed node groups and managed networking add-ons will no longer be created by default.
2026-05-27 20:59:57 [i] Auto ModeConfig.enabled: false' in your cluster configuration. Learn more: https://eksctl.io/usage/auto-mode/
2026-05-27 20:59:57 [i] using Kubernetes version 1.34
2026-05-27 20:59:57 [i] creating EKS cluster "b17d-eks-cluster" in "us-east-1" region with managed nodes
2026-05-27 20:59:57 [i] will create 2 separate CloudFormation stacks for cluster itself and the initial managed nodegroup
2026-05-27 20:59:57 [i] if you encounter any issues, check CloudFormation console or try 'eksctl utils describe-stacks --region=us-east-1 --cluster=b17d-eks-cluster'
2026-05-27 20:59:57 [i] Kubernetes API endpoint access will use default of {publicAccess=true, privateAccess=false} for cluster "b17d-eks-cluster" in "us-east-1"
2026-05-27 20:59:57 [i] Cloudwatch logging will not be enabled for cluster "b17d-eks-cluster" in "us-east-1"
2026-05-27 20:59:57 [i] you can enable it with 'eksctl utils update-cluster-logging --enable-types={SPECIFY-YOUR-LOG-TYPES-HERE (e.g. all)} --region=us-east-1 --cluster=b17d-eks-cluster'
2026-05-27 20:59:57 [i] default add-ons metrics-server, vpc-cni, kube-proxy, coredns were not specified, will install them as EKS add-ons
2026-05-27 20:59:57 [i]
2 sequential tasks: { create cluster control plane "b17d-eks-cluster",
  2 sequential sub-tasks: {
    2 sequential sub-tasks: {
      1 task: { create addons },
      wait for control plane to become ready,
    },
    create managed nodegroup "b17d-nodegroup",
  }
}
2026-05-27 20:59:57 [i] building cluster stack "eksctl-b17d-eks-cluster-cluster"
2026-05-27 20:59:59 [i] deploying stack "eksctl-b17d-eks-cluster-cluster"
2026-05-27 21:00:29 [i] waiting for CloudFormation stack "eksctl-b17d-eks-cluster-cluster"
2026-05-27 21:01:00 [i] waiting for CloudFormation stack "eksctl-b17d-eks-cluster-cluster"
2026-05-27 21:02:02 [i] waiting for CloudFormation stack "eksctl-b17d-eks-cluster-cluster"
```

```
2026-05-27 21:02:02 [i]
2 sequential tasks: { create cluster control plane "b17d-eks-cluster",
  2 sequential sub-tasks: {
    2 sequential sub-tasks: {
      1 task: { create addons },
      wait for control plane to become ready,
    },
    create managed nodegroup "b17d-nodegroup",
  }
}
2026-05-27 20:59:57 [i] building cluster stack "eksctl-b17d-eks-cluster-cluster"
2026-05-27 20:59:59 [i] deploying stack "eksctl-b17d-eks-cluster-cluster"
2026-05-27 21:00:29 [i] waiting for CloudFormation stack "eksctl-b17d-eks-cluster-cluster"
2026-05-27 21:01:00 [i] waiting for CloudFormation stack "eksctl-b17d-eks-cluster-cluster"
2026-05-27 21:02:02 [i] waiting for CloudFormation stack "eksctl-b17d-eks-cluster-cluster"
2026-05-27 21:03:03 [i] waiting for CloudFormation stack "eksctl-b17d-eks-cluster-cluster"
2026-05-27 21:04:06 [i] waiting for CloudFormation stack "eksctl-b17d-eks-cluster-cluster"
2026-05-27 21:05:07 [i] waiting for CloudFormation stack "eksctl-b17d-eks-cluster-cluster"
2026-05-27 21:06:08 [i] waiting for CloudFormation stack "eksctl-b17d-eks-cluster-cluster"
2026-05-27 21:07:09 [i] waiting for CloudFormation stack "eksctl-b17d-eks-cluster-cluster"
2026-05-27 21:08:11 [i] waiting for CloudFormation stack "eksctl-b17d-eks-cluster-cluster"
2026-05-27 21:09:12 [i] waiting for CloudFormation stack "eksctl-b17d-eks-cluster-cluster"
2026-05-27 21:10:13 [i] waiting for CloudFormation stack "eksctl-b17d-eks-cluster-cluster"
2026-05-27 21:11:14 [i] waiting for CloudFormation stack "eksctl-b17d-eks-cluster-cluster"
2026-05-27 21:12:16 [i] waiting for CloudFormation stack "eksctl-b17d-eks-cluster-cluster"
2026-05-27 21:13:17 [i] waiting for CloudFormation stack "eksctl-b17d-eks-cluster-cluster"
2026-05-27 21:13:23 [i] recommended policies were found for "vpc-cni" addon, but since OIDC is disabled on the cluster, eksctl cannot configure the requested permissions; the recommended way to provide IAM per
missions for "vpc-cni" addon is via pod identity associations; after addon creation is completed, add all recommended policies to the config file, under 'addon.PodIdentityAssociations', and run 'eksctl update a
addon'
2026-05-27 21:13:23 [i] creating addon: vpc-cni
2026-05-27 21:13:23 [i] successfully created addon: vpc-cni
2026-05-27 21:13:24 [i] creating addon: kube-proxy
2026-05-27 21:13:25 [i] successfully created addon: kube-proxy
2026-05-27 21:13:26 [i] creating addon: coredns
2026-05-27 21:13:26 [i] successfully created addon: coredns
2026-05-27 21:15:36 [i] building managed nodegroup stack "eksctl-b17d-eks-cluster-nodegroup-b17d-nodegroup"
2026-05-27 21:15:39 [i] deploying stack "eksctl-b17d-eks-cluster-nodegroup-b17d-nodegroup"
2026-05-27 21:15:39 [i] waiting for CloudFormation stack "eksctl-b17d-eks-cluster-nodegroup-b17d-nodegroup"
2026-05-27 21:16:10 [i] waiting for CloudFormation stack "eksctl-b17d-eks-cluster-nodegroup-b17d-nodegroup"
2026-05-27 21:16:52 [i] waiting for CloudFormation stack "eksctl-b17d-eks-cluster-nodegroup-b17d-nodegroup"
2026-05-27 21:17:55 [i] waiting for CloudFormation stack "eksctl-b17d-eks-cluster-nodegroup-b17d-nodegroup"
2026-05-27 21:43:35 [i] waiting for CloudFormation stack "eksctl-b17d-eks-cluster-nodegroup-b17d-nodegroup"
2026-05-27 21:43:35 [i] 1 error(s) occurred and cluster hasn't been created properly, you may wish to check CloudFormation console
2026-05-27 21:43:35 [i] to cleanup resources, run 'eksctl delete cluster --region=us-east-1 --name=b17d-eks-cluster'
2026-05-27 21:43:35 [x] expected err to be of type smithy.APIError, got operation error CloudFormation: DescribeStacks, https response error StatusCode: 0, RequestID: , canceled, context deadline exceeded
Error: failed to create cluster "b17d-eks-cluster"
```

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/OneDrive/Desktop
$
```

Amazon Elastic Kubernetes Service

Dashboard

Clusters

Settings

Amazon EKS Anywhere

Related services

Documentation

EKS workshops

b17d-eks-cluster

ConnectDelete clusterUpgrade versionMonitor cluster

End of standard support for Kubernetes version 1.34 is December 2, 2026.

Cluster info

StatusActive

Cluster health0

Kubernetes version1.34

Upgrade insights5

Support periodStandard support until December 2, 2026

Node health issues0

ProviderEKS

Capability issues0

OverviewResourcesComputeNetworkingAdd-ons1CapabilitiesAccessObservabilityUpdate history & backupsTags

Nodes (2)

Filter Nodes by property or value

Node name	Instance type	Compute	Managed by	Created	Status	CPU usage	Memory usage	Ephemeral storage usage
ip-192-168-28-129.ec2.internal	t3.micro	Node group	b17d-nodegroup	31 minutes ago	Ready	22%	59%	11%
ip-192-168-35-246.ec2.internal	t3.micro	Node group	b17d-nodegroup	31 minutes ago	Ready	12%	44%	11%

Node groups (1)

Node groups implement basic compute scaling through EC2 Auto Scaling groups.

Filter node groups by property or value

Group name	Desired size	AMI release version	Launch template	Status
b17d-nodegroup	2	1.34.8-20260520	eksctl-b17d-eks-cluster-nodegroup-b17d-nodegroup (1)	Active

Fargate profiles (0)

Filter Fargate profiles by property or value

us-east-1.console.aws.amazon.com/cloudformation/home?region=us-east-1#/stacks?filteringText=&filteringStatus=active&viewNested=true

CloudFormation

Stacks

Stack refactorsStackSetsExportsInfrastructure ComposerIaC generatorHooks overviewInvocation summaryHooks

Stacks (2)

Search by stack name

Filter statusActiveView nested

Stack name	Status	Created time	Description
eksctl-b17d-eks-cluster-nodegroup-b17d-nodegroup	CREATE_COMPLETE	2026-05-27 21:15:39 UTC+0530	EKS Managed Nodes (SSH access: false) [created by eksctl]
eksctl-b17d-eks-cluster-cluster	CREATE_COMPLETE	2026-05-27 20:59:59 UTC+0530	EKS cluster (dedicated VPC: true, dedicated IAM: true) [created and managed by eksctl]

VPC

Your VPCs

VPC dashboardAWS Global ViewFilter by VPC

Virtual private cloud

Your VPCs

SubnetsRoute tablesInternet gatewaysEgress-only internet gatewaysCarrier gatewaysDHCP option setsElastic IPsManaged prefix listsNAT gatewaysPeering connectionsRoute servers

Security

PrivateLink and Lattice

Getting startedEndpointsEndpoint servicesService networks

Your VPCs

VPCsVPC encryption controls

Your VPCs (1/2)

Find VPCs by attribute or tag

Name	VPC ID	State	Encryption c...	Encryption control ...	Block Public...	IPv4 CIDR	IPv6 CIDR
eksctl-b17d-eks-cluster-cluster/VPC	vpc-0da75986ab21a618f	Available	-	-	Off	192.168.0.0/16	-
-	vpc-023934cbcb5b6f36	Available	-	-	Off	172.31.0.0/16	-

vpc-0da75986ab21a618f / eksctl-b17d-eks-cluster-cluster/VPC

Resource map

VPC

Your AWS virtual network

eksctl-b17d-eks-cluster-cluster/VPC

Subnets (4)

Subnets within this VPC

us-east-1b

eksctl-b17d-eks-cluster-cluster/SubnetPublicUSEAST1B

eksctl-b17d-eks-cluster-cluster/SubnetPrivateUSEAST1B

us-east-1c

eksctl-b17d-eks-cluster-cluster/SubnetPublicUSEAST1C

eksctl-b17d-eks-cluster-cluster/SubnetPrivateUSEAST1C

Route tables (4)

Route network traffic to resources

rtb-05af0a00584e489d0

eksctl-b17d-eks-cluster-cluster/PrivateRouteTableUSEAST1C

eksctl-b17d-eks-cluster-cluster/PublicRouteTable

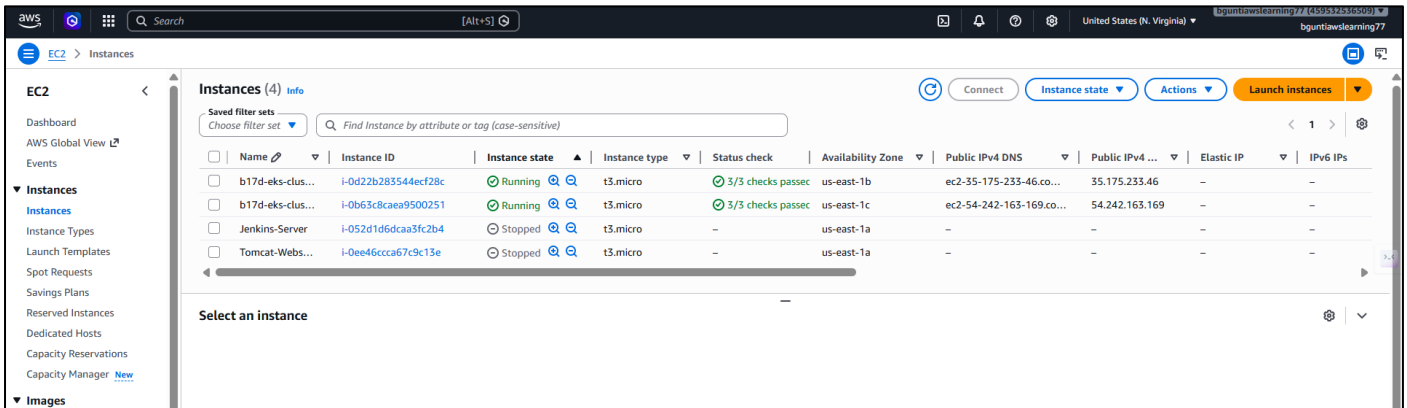
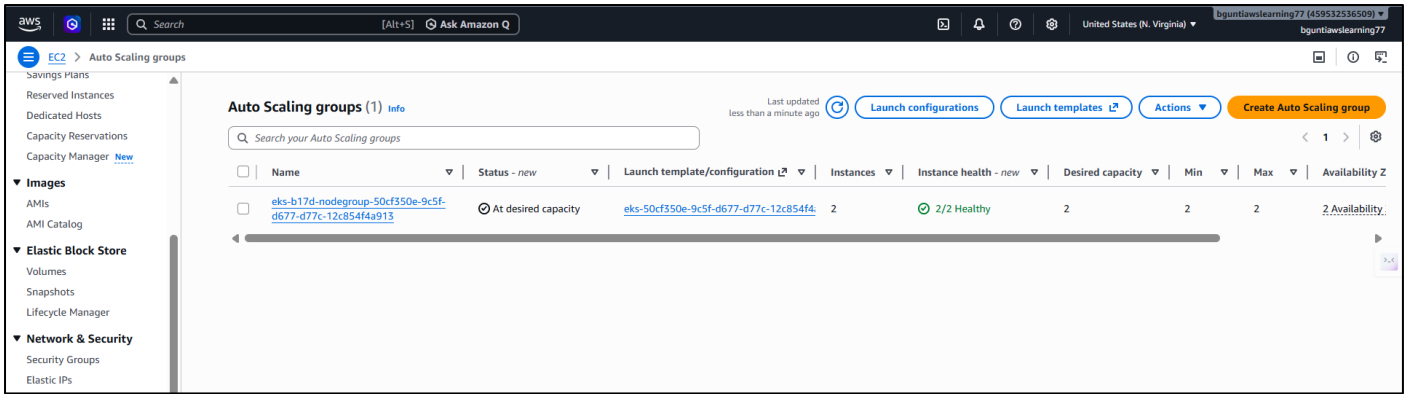
eksctl-b17d-eks-cluster-cluster/PrivateRouteTableUSEAST1B

Network Connections (2)

Connections to other networks

eksctl-b17d-eks-cluster-cluster/InternetGateway

eksctl-b17d-eks-cluster-cluster/NATGateway



4. Commands to get cluster, nodes, create pod inside nodes, delete eks-cluster

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/OneDrive/Desktop
$ eksctl get cluster
```

```
NAME          REGION    EKSTCL CREATED
b17d-eks-cluster  us-east-1  True
```

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/OneDrive/Desktop
$ kubctl get nodes
```

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/OneDrive/Desktop
$ kubctl run testpod1 --image=nginx
```

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/OneDrive/Desktop
$ kubectl get pods
```

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/OneDrive/Desktop
$ kubectl get po
```

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/OneDrive/Desktop
$ kubectl get po
```

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/OneDrive/Desktop
$ kubectl desc testpod1
```

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/OneDrive/Desktop
$ kubectl get pods
```

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/OneDrive/Desktop
$ eksctl get cluster
```

```
NAME          REGION    EKSTCL CREATED
b17d-eks-cluster  us-east-1  True
```

```
Bhavana@LAPTOP-E0A1LIK MINGW64 ~/OneDrive/Desktop
$ eksctl delete b17d-eks-cluster
```